MILITARY SPECIFICATION

# FILTER ELEMENT, AGENT SAMPLING EXTRACTOR

This limited coordination military specification has been prepared by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA, 19101 but it has not been approved for promulgation as a coordinated military specification. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

## 1. SCOPE

1.1 This specification covers a 1-7/8 inch diameter filter disk (element) for use in sampling air for the presence of radioactive particulates.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-566 | - Boxes, Folding, Paperboard. |
| PPP-B-636 | - Box, Fiberboard. |
| PPP-B-676 | - Boxes, Set-Up, Paperboard. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640



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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

136-300-138B - Instruction Manual for the Installation, Operation and Maintenance of Penetrometer, Filter Testing, DOP, Q127

(Application for copies should be addressed to the Production Assurance Directorate, Edgewood Arsenal, Aberdeen Proving Ground, MD 21010.)

3. REQUIREMENTS

3.1 Filter material. The filter material shall be made from either fiberglass strands and microglass fibers, or microglass fibers that are flexibly bound with acrylic resin. The filter material shall have physical properties conforming to table I, when tested as specified in the 4.4.1.

TABLE I. Physical Properties of Filter Material

| Characteristic | Requirement |
|---|---|
| Basis weight per 3000 square feet | 37 ± 5 pounds |
| Dry thickness | 0.014 ± 0.003 inches |
| Tensile strength | |
| Machine direction | 2.6 pounds per inch width (min.) |
| Cross direction | 2.0 pounds per inch width (min.) |

3.2 Filter material performance characteristics.

3.2.1 Air resistance. The air resistance for a single layer of filter material shall not exceed 23.6 millimeters of mercury with an air velocity of 200 centimeters per second passing through the specimen, when tested as specified in 4.4.2.

3.2.2 Dioctyl phthalate (DOP) penetration. The DOP penetration for a single layer of filter material shall be less that 10 percent using a 0.3 micron DOP aerosol in an airstream passing through the specimen a velocity of 200 centimeters per second, when tested as specified in 4.4.2.

3.3 Construction of filter element. The filter element shall be a single layer in the form of a 1-7/8 inch diameter disk, with +1/16 and -0 tolerances on the dimension. The downstream face of the filter disk (i.e., with wireside impression) shall have either a 5/16 to 7/16 inch diameter red dot or a 1/16 to 1/8 inch wide red line, that is applied with a water soluble dye having an affinity for the filter material. The dye mark shall not restrict the flow of air through the filter element.

3.4 Workmanship. The filter element shall be free of defects that detract from its appearance or impair its serviceability.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. A certificate of quality supplied by the manufacturer of the filter material may be furnished in lieu of actual performance of such testing by the contractor. The certificate shall include the name of the manufacturer or supplier, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and specification requirements. The test method shall include a brief description of equipment and test methods used, and shall also include any reference to applicable industrial standards (e.g., Technical Association for the Pulp and Paper Industry). The contractor shall provide the ordering activity with a copy of the certificate and shall include a statement that only filter material covered by the certificate was used in making the filter disks.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105 and table II. The unit of product for inspection purposes shall be the unit package (see 5.1). All samples shall be randomly selected. One filter element from each sample unit shall be examined for defects.

TABLE II. Sampling for examination

| | Inspection level | Acceptance Quality Level (percent defective) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 1.0 |
| Minor defects | II | 4.0 |
| For dimensional examination (diameter only) | II | 1.0 |
| For the count | S-2 | 2.5 |



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4.2.2 For tests. Sampling for tests shall be conducted in accordance with table III. The unit of product for testing shall be each slit roll of filter material that is used in the fabrication of the disks. The contractor shall make arrangements with the manufacturer of the filter material for selecting test strip samples from each roll, in accordance with the following:

1. One three-foot test strip sample selected from the free end of the each slit roll at a sufficient distance from the edge of the roll to preclude inclusion of material damaged due to handling, wrapping and storage.

2. One three-foot test strip selected from the middle of the roll.

3. One three-foot test strip selected from the core end of the roll.

If the manufacturer supplies the filter material in the form of flat stock, the requirements of this paragraph shall be applicable to each slit roll used in making the flat stock. Each test strip shall be suitably identified so as to be traceable to the slit roll from which it was taken.

TABLE III. Sampling for tests.

| Characteristic | Requirement | Test procedure | Inspection level | AQL |
|---|---|---|---|---|
| Basis weight | 3.1 | 4.4.1 | * | ** |
| Dry thickness | 3.1 | 4.4.1 | * | ** |
| Tensile strength | 3.1 | 4.4.1 | * | ** |
| Air resistance | 3.2.1 | 4.4.2 | * | ** |
| DOP penetration | 3.2.2 | 4.4.2 | * | ** |

* See requirements of this paragraph.
** Acceptance number shall be zero.

4.3 Examination: The filter elements shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table IV. Examination shall not be restricted to the possible defects listed in the table.

TABLE IV. Classification of defects

| Categories | Defects |
|---|---|
| Major | |
| 101 | Filter element not free of tears, cuts or punctures. |
| 102 | Filter element not free of abrasion. |
| 103 | Filter element does not include red mark on the wire side. |
| 104 | Filter element not free of foreign material. |
| Minor | |
| 201 | Red mark dimension on filter element not conforming to requirements but readily identifiable. |
| 202 | Filter element not free of faint bleed-through of dye from wire side. |



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4.3.2 Dimensional examination. The diameter of the filter element shall be examined for a defect in the dimension. Any dimension not within the specified limit shall be classified as a defect.

4.3.3 Count. The contents of a unit package (see 5.1) shall be checked for a sufficient quantity of filter elements. Any unit package not containing at least 10 filter elements shall be classified as a defect.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging and marking requirements of Section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 percent defective.

TABLE V. Examination of preparation for delivery

| Categories | Defects |
|---|---|
| Container | Not as specified. |
| Contents | Not as specified. |
| Marking | Omitted, incorrect, illegible, improper size, location, sequence or method of application. |
| Packaging and packing materials | Component missing or damaged. |
| Workmanship | Bulging or distortion of containers. Cushioning inadequate or improper. |

4.4 Tests. Tests shall be conducted to determine compliance with this specification.

4.4.1 Basis weight, thickness and tensile strength. The physical properties of the filter material shall be determined in accordance with test procedures and equipment normally used by the manufacturer for that purpose.

4.4.2 Air resistance and DOP penetration. The air resistance and DOP penetration shall be determined for the same single layer specimen held in an orifice plate fixture that is constructed in accordance with figure 1 of the appendix in this specification, and tested in a Q-127 DOP Penetrometer that is equipped to measure air velocity in centimeters per second and air resistance in millimeters of mercury. The procedure specified in Instruction Manual 136-300-138B shall apply except for the use of the fixture, a mercury manometer (or equivalent) and flow control at 200 centimeters per second. As indicated in the appendix, the orifice plate fixture shall be a two-plate fixture consisting of:

a. An upstream plate with a 1.5 centimeter diameter opening.
b. A downstream plate with a 1.5 centimeter diameter opening.
c. A stainless steel screen (40x40 mesh, 0.0085 inch wire, with 0.0165 inch opening) that is mounted in the downstream plate's opening flush with the plate's upstream face.



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The filter material specimen shall be of sufficient size to be installed between the plates and cover the opening and pilot pins in the downstream plate. The filter material shall be installed with its wire side against the screen.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing ten filter elements, as specified, constitutes a unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table VI shall be packaged in each unit, intermediate and exterior container as applicable for the required level of protection specified in the procurement document.

TABLE VI. Packaging quantities.

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 10 units | 1000 units |

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Ten filter elements shall be clean, dry and packaged in a sealed peel-open type envelope of appropriate size and design.

5.1.3.2 Intermediate package. The intermediate package shall be a box conforming to PPP-B-566 or PPP-B-676.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the box specification.



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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Filter material.

6.2.1 Intended use. The filter element is intended for use in portable air samplers to measure airborne particulate radioactivity in radiological work areas.

6.3 Prospective material. Manufacturers usually test their filter material with 0.3 micron dioctyl phthalate (DOP) aerosol at 5.3 centimeters per second (32 liters per minute through a 100 square centimeter specimen). Filter material that has been tested in this manner and found to have DOP penetration of 13 $\pm$ 8 percent and an air resistance of 8 $\pm$ 2 millimeters of water, and which meets the requirements of 3.1, might meet the requirements of 3.3.1 and 3.3.2.

6.4 This specification covers the following item listed in the Federal Supply Catalog:

| National Stock Number | Item Identification |
|---|---|
| 6640-01-138-5753 | FILTER ELEMENT, AGENT SAMPLING EXTRACTOR |

Custodians:
Army - EA
Navy - SH
Air Force - 99

Preparing Activity:
DoD-DM

Project No. 6640-1219



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# ORIFICE PLATE FIXTURE

## GENERAL DESCRIPTION

The orifice plate fixture herein is intended for use with the Air Techniques, Incorporated (Baltimore, MD) Model Q127 DOP Penetrometer. The purpose of the fixture is to hold a specimen of filter material while an airstream having 0.3 micron monodispersed dioctyl phthalate aerosol dispersed therein passes through the filter at a velocity of 200 centimeters per second.

The fixture consists of two, 1/8 inch thick brass plates of a diameter compatible for retention in the chuck of the penetrometer (expected to be about 5 inches). The upstream plate (called "top plate" in figure 1) has a 1.5 cm diameter center opening and two index holes to receive the index pins of the downstream plate. The pins prevent the plates from shifting.

The downstream (or bottom) plate has a 1.5 cm diameter center opening that contains a stainless steel screen soldered in place so that it is on the same plane as the plate's upstream surface. Two index pins protrude from this plate to penetrate the sheet of filter specimen and mate with the index holes of the upstream plate.

Each plate has a scribe mark on the outer edge so that the upstream plate can be aligned in the same position with the downstream plate each time the fixture is used. A rubber gasket on each plate will provide a seal with the chuck of the penetrometer.

## MATERIALS

1. Plates - brass 1/8" thick with 1.5 cm diameter center hole.
2. Index pins - brass about 3/32" diameter by 5/32" long.
3. Screen - stainless steel 40x40 mesh .0085" wire with .0165" opening.
   For example: Newark Wire Cloth Company Telegraph Code Type ZXA.
4. Solder - tin or equivalent.
5. Gasket - oil resistant rubber.

## FINAL ASSEMBLY

When plates are aligned by the index pins and scribe marks and are in contact with each other:

1. No shifting should occur.
2. No rocking of one plate on the other should occur.
3. The center holes should line up.
4. The solder should not reduce the 1.5 cm opening of the downstream plate.



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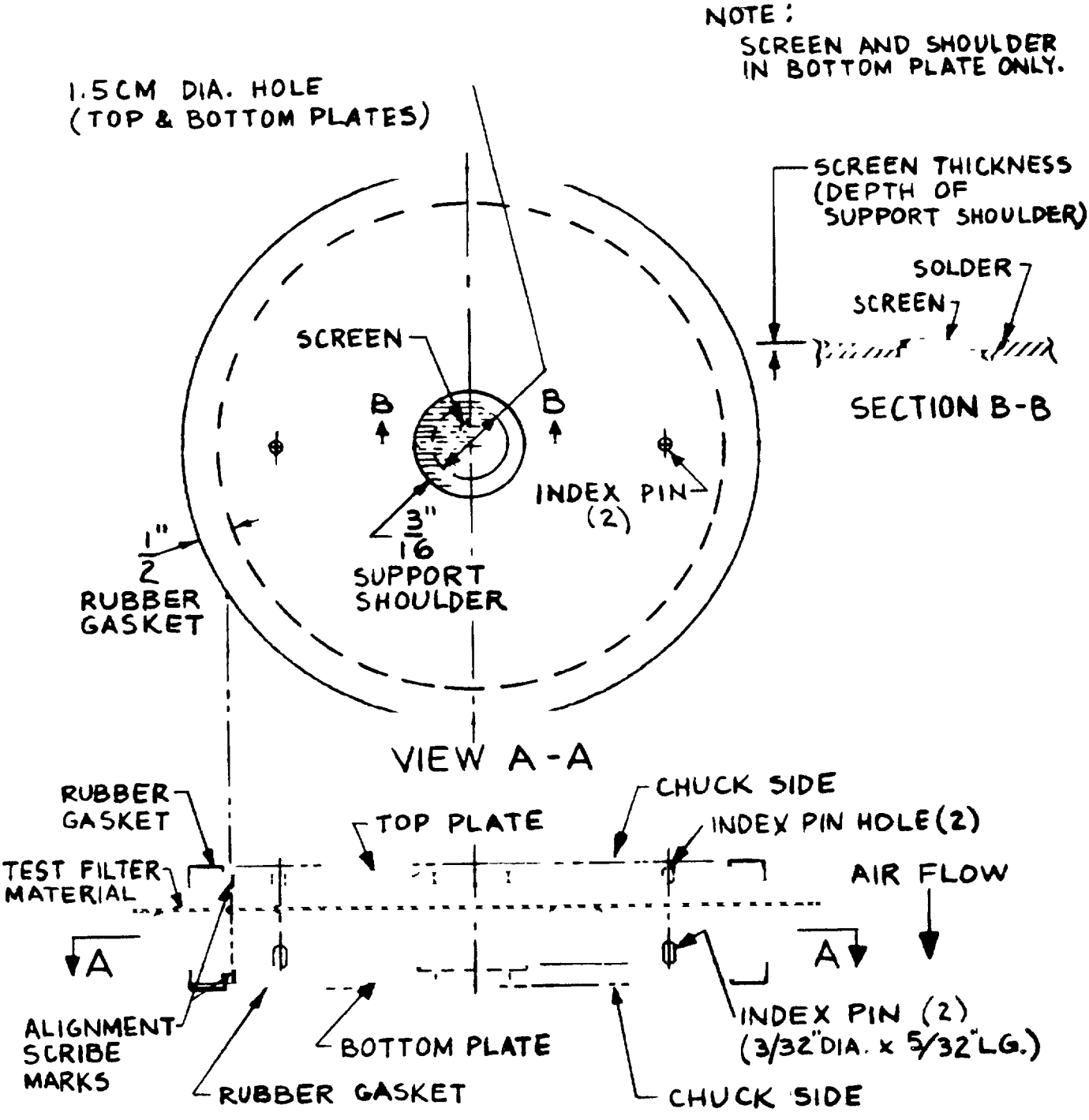



FIGURE 1 - 1.5CM ORIFICE PLATE FIXTURE FOR DOP TEST UNIT



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

1. DOCUMENT NUMBER: MIL-F-37956 (DLA-DM)

2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION

b ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional

b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c MAILING ADDRESS *(Street City State ZIP Code)* – Optional

8 DATE OF SUBMISSION *(YYYYMMDD)*

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